

PracticePro

Architecture & Context Reference

PLATFORM DESIGN DOCUMENT

For NotebookLM Design Reference

PracticePro Legal Technologies Ltd | Lagos, Nigeria
NDPA 2023 Compliant | ISO 27001 Aligned | AES-256 Encrypted

1. Product Overview & Architecture

PracticePro is a multi-product SaaS platform serving Nigerian legal and property management professionals. The platform operates as three distinct products under a unified codebase, each targeting a specific market segment while sharing common infrastructure, authentication, and data models.

Three Products

Product	Market	Description
Vega	Legal Practice	Matter management, document drafting, time tracking, billing, client portal
Atrium	Property Management	Property/unit management, rent collection, service charge tracking, resident port
Complete	Unified Practice	Hybrid of Vega + Atrium for firms that serve both legal and property clients

Technology Stack

Backend: Convex (serverless, real-time, TypeScript). Schema-defined tables with indexes. Mutations, queries, actions, cron jobs, HTTP endpoints. Custom auth (email/password, PBKDF2-SHA512, 600k iterations).

Frontend: React 18 + TypeScript + Vite. Tailwind CSS 4. Convex React hooks (useQuery, useMutation). Lucide icons. Custom modal system with portal rendering.

Mobile: Capacitor (Android APK). Two separate APKs: consumer app + admin/founder app. GitHub Actions CI/CD for automated APK builds.

Deployment: Cloudflare Workers (primary, practiceprovega.prototypechigo.workers.dev). Vercel (backup, practice-pro-vega.vercel.app). Convex backend deployed via CLI.

Integrations: Brevo (transactional email), Chakra Chat API (WhatsApp), Paystack (payment gateway), Google Drive, Gemini AI (document drafting).

2. Admin (Founder) APK Structure

The founder app is a separate Capacitor APK with its own Vite build config, entry point (admin.html), and bottom navigation. It provides platform-level oversight across all firms, with privacy controls that prevent the founder from seeing individual portal user (resident/client) PII.

Views & Navigation

View	Purpose
Dashboard	KPIs (MRR, total users, firms, active trials), action items, trial funnel
Alerts	Unified notifications center: signups, security events, subscriptions, churn, milestones. Filter tabs, mark-as-read
Organizations	Firm management table with presence glow (emerald for online firms), master-detail drawer, plan editing, trial
Feedback	User feedback submissions inbox
Security	Live session feed (who's online), active firms with emerald tint, security alerts panel (disposable emails, unau
Subscriptions	Plan upgrade request approval queue with discount slider
Broadcast	Push notification/broadcast console to firms
Audit	Generic audit trail (signups, plan changes, role changes)
Settings	Account, appearance, security toggles, notifications, system, about
System	Runtime error feed (Gemini failures, PDF errors)
Export	CSV data export tools
Analytics	Growth chart, active users, MRR breakdown

Badge System

The Alerts nav item shows two badges: **red** for pending subscription requests, **emerald** for new firm team signups (Admin/Lawyer/Paralegal only, not portal users). Both use formatBadgeCount() which caps at '9+'. The More button shows a red dot when pending requests exist.

Presence Engine

The Organizations table polls getAllPresenceForAdmin every 30s. Firms with at least one online user get: emerald background tint (rgba(16,185,129,0.08)), 4px left border (#10b981), pulsing emerald dot (animate-pulse). The Security Center shows a live session feed with all online users.

3. User APK Structure

The consumer app serves firm users (Admin, Lawyer, Paralegal) with role-based feature access. The app adapts its terminology, navigation, and feature set based on the firm's product (Vega/Atrium/Komplete).

Core Views

Properties (Atrium/Komplete): Property list, property detail with 5 tabs (Summary, Units, Notices, Financials, Activity & Tracking). Unit cards with service charge/minimum vend progress bars, Quick Payment Drawer, receipt pipeline, resident contact auto-sync.

Matters (Vega/Komplete): Matter list/board, matter detail with accordion (Practice Area, Matter Title, Client, Assigned Team, Billing, Case Details), AI-assisted document drafting (ALOA).

Other views: Contacts, Documents, Tasks (with Priority/Status inline dropdowns), Calendar, Billing (invoices, time entries, expenses, retainer), Messages (team chat + portal conversations + inbound WhatsApp/Email), Reports, Settings.

Property Unit Cards

Each unit card shows: status border, resident name, lease end, service charge status pills (green=paid on time, orange=paid late, red=outstanding, blue=advance paid). Buttons: Message, Edit, More. Expanded view shows: SC/MV progress bars with per-period pills, payment history, maintenance tickets, eviction tracker, statutory timeline milestone.

Service Charge Progress System

The ServiceChargeBars component renders per-period status pills computed from leaseStart + billing frequency. Calendar-month arithmetic (not fixed 30.44 days). SC uses its own serviceChargeFrequency (defaults to rentFrequency). OnboardUnitLedgerModal provides bulk-settle controls (Mark All Paid On Time / Paid Late / Reset All). Quick Payment Drawer allows per-period status toggling with auto-receipt generation (zero-touch: marks paid, creates receipt, publishes to resident portal, writes activity log, updates ledger — all in one click).

4. Resident Portal Experience

The Resident Portal is a separate web experience (not the APK) accessed via browser at `/portal/tenant/`. Residents log in with their email + password (set up via the invite flow). The portal is branded as 'Atrium' for property residents and 'Vega' for legal clients, regardless of the firm's *Komplete* status.

Portal Invite Flow

1. Admin clicks [Send Portal Invite] on a unit card or in `ComposeMessageModal`.
2. Backend (`createPortalInvite` action) generates a secure token (7-day expiry), creates a `portal_invites` record, and sends a clean invite email via Brevo.
3. Email contains ONLY: greeting, portal description, 'Set Up My Password' button, fallback link, 7-day expiry notice. No admin message in the email.
4. Resident clicks button, lands on `/setup-password?token=xxx`, sees locked email field, creates password, accepts terms, activates account.
5. Admin's message (if any) appears as an in-app portal message AFTER login, not in the email.

Portal Tabs

Tab	Features
Dashboard	Overview cards, quick stats, recent activity
Notices	Broadcast notices from property management
Ledger	Payment history, outstanding balances, service charge status
Receipts	Auto-generated PDF receipts for SC/MV payments
Documents	Shared documents (lease agreements, terms, consent records)
Messages	Two-way messaging with property manager (same conversation thread)
Maintenance	Submit maintenance requests with priority levels
Visitors	VMS — generate visitor access codes (if enabled)

Portal Messaging

PM to resident messages use `sendPortalMessage` with `admin→resident participantId` resolution. The backend resolves the resident's `userId` from the `unitId` (`property → rentalDetails.tenantEmail → users` table). Sets `unreadByParticipant` (not `unreadByAdmin`) so the resident sees a badge. Resident replies use the same `conversationId` — no separate threads. Notification badges show on the Messages tab.

5. Design System & Patterns

Color Palette

Token	Hex	Usage
Primary-600	#0ea5e9	Primary actions, links, active states, brand accent
Emerald-500	#10b981	Success, paid on time, online presence, portal active
Amber-500	#f59e0b	Warnings, paid late, churn risk, pending requests
Red-500	#ef4444	Errors, outstanding/overdue, security alerts
Blue-500	#3b82f6	Advance paid, rent collection, information
Violet-500	#8b5cf6	Assigned team, portal invite, subscriptions
Slate-900	#0f172a	Dark backgrounds, headings, portal dark theme
Slate-50	#f8fafc	Light backgrounds, card surfaces

Typography

Inter (sans-serif) for all UI text. Font sizes: text-2xs (10px), text-3xs (9px) for labels/badges, text-xs (12px) for body, text-sm (14px) for emphasized content, text-lg/xl for headings. Font weights: font-bold (700) for labels, font-black (900) for section headings. Uppercase tracking-widest for category labels.

Component Patterns

AccordionSection: Module-level `React.memo` component with colored icon badge, expand/collapse animation, `scrollIntoView` on expand. Used in `PropertyForm` (6 sections) and `MatterForm` (6 sections).

StatCard: Uniform height (h-28), watermark icon top-right, marquee text for overflow, tooltip for full values. Used in `PropertyDetailView` Financials tab.

Modals: Portal-rendered (`createPortal` to `document.body`) with `bg-black/60` backdrop, `z-[9990]+`. Includes: `ComposeMessageModal`, `ReceiptModal`, `OnboardUnitLedgerModal`, `QuickPaymentDrawer`, `CriticalLeaseBanner`, `TaskDetailModal`, `ComposeEmailModal`.

Progress Pills: Slim h-2 w-7 rounded-full color-only bars (no text). Portal-rendered tooltips with month/year, amount, status, settled date. Click opens Quick Payment Drawer.

Toasts: Max 3 visible, 4s auto-dismiss, positioned bottom-center. Types: success (emerald), error (red), info (blue).

6. Security Infrastructure

Implemented Security Measures

Measure	Status	Details
Rate Limiting	Active	5 signups/email/minute via rateLimits table. Violations logged to securityEvents.
Disposable Email Blocking	Active	32 known disposable domains blocked on signup. Logged as security events.
RLS Audit Logging	Active	requireFirmUser logs unauthorized_access events. Suspended users checked.
Password Hashing	Active	PBKDF2-SHA512, 600k iterations, 16-byte salt. Legacy SHA-256 auto-migration.
Portal Invite Security	Active	7-day token expiry, participantId tied to email, revoke/suspend/delete paths.
Presence Engine	Active	20s heartbeat, 90s online threshold. Presence privacy flag for portal users.
Admin Actions	Active	blockIp, unblockIp, suspendUser, unsuspendUser mutations (founder-gated).
Bot Detection	Planned	HTTP middleware inspecting user-agent strings
Impossible Travel	Planned	IP geolocation logging on login
FCM/APNs Push	Planned	Native push notification engine for mobile

Privacy Controls

The founder app filters portal users (Tenant, Client, Resident) from all alerts, badges, and signup counts. Only firm team members (Admin, Lawyer, Paralegal) appear in founder-facing data. Portal user activity is tracked at the org/firm level via presence and engagement metrics, but individual portal user PII (names, emails) is never exposed to the founder.

7. Known Gaps & Improvement Opportunities

High Priority

1. **Uniform conversation styling:** MessagesView has 5 different card layouts (System Inbox, Inbound WA/Email, Team DM, Portal, Legacy Client). Need unified structure: avatar + name + type badge + preview.
2. **FCM/APNs push notifications:** No native push engine. Residents don't get push notifications for new portal messages, receipts, or notices. Requires Firebase Cloud Messaging integration.
3. **Custom estate fees + itemized receipts:** Revenue Breakdown table only supports Rent, SC, and MV. Need dynamic custom fee categories with per-period tracking and itemized PDF receipts.

Medium Priority

4. **Headless bot detection:** HTTP middleware inspecting user-agent strings for Puppeteer/Playwright/Selenium/cURL signatures.
5. **Impossible travel detection:** Log IP + geo on login, flag accounts that log in from vastly different locations within short intervals.
6. **26 orphaned dead code files:** Legacy components that are imported but unused. Should be cleaned up to reduce bundle size and maintenance burden.
7. **Founder notifications center improvements:** Add broadcast delivery reports, one-click security actions (Suspend/Block IP buttons), notification preferences.

Design Polish

8. **Mobile layout optimization:** Several views still require scrolling on mobile (setup-password page fixed, but others may need condensing).
9. **Onboarding wizard:** New firm onboarding could be more guided with progress indicators and contextual help.
10. **Dark mode consistency:** Some components have incomplete dark mode variants (zinc-800/zinc-900 classes missing in places).

8. Recent Work Summary (This Session)

This document was generated after an extensive development session covering UI refactors, security infrastructure, portal messaging fixes, and admin app enhancements. Below is a summary of what was built and fixed.

Area	What Was Done
MatterForm	Accordion layout (6 sections), pinned footer, keyboard focus auto-expand, assigned team section
PropertyForm	Accordion layout, auto-scroll to rental section, resident email field, save-to-contacts checkbox, decoupled property/U
Service Charge Bars	Per-period colored pills (green/orange/red/blue), calendar-month stepping, SC frequency independent from rent, bu
Receipt Pipeline	Zero-touch auto-receipt on payment: generates receipt, publishes to resident portal, writes activity log, updates ledg
Quick Payment Drawer	Portal-rendered, historical pill strip, 4-state toggle (Paid On Time / Paid Late / Outstanding / Advance Paid), pointer
Portal Invite	Fixed: now calls createPortalInvite (sends real email via Brevo) instead of sendPortalMessage (just DB row). Clean
Portal Messaging	Fixed: admin messages now reach residents (participantId resolved from unitId→tenantEmail→userId). Correct untr
Resident Portal	Setup-password page condensed to fit one screen, correct tab title ('Atrium — Residents' Portal')
Banner System	Swipe carousel (touch handlers), per-notification dismissal (notifId-keyed, not broadcastId-keyed)
Admin App	SecurityCenter view, FounderNotificationsCenter, presence glow in OrganizationsHub, signup badge (9+ cap), Org
Security	Rate limiting (rateLimits table), disposable email blocking (32 domains), RLS audit logging, blockIp/suspendUser m
Terminology	Tenant → Resident across all UI labels (field names unchanged for data compatibility)
Mobile	Touch handlers (onTouchEnd) on all unit card buttons, safe-area padding on property overview

This document is intended as a reference source for NotebookLM. When combined with a UI design video, it should enable the generation of targeted improvement prompts for both the admin and user APKs. The document covers the current state of the platform, known gaps, and the design system conventions that should be followed in any future UI work.